

LARAMIE, WY, USA

WMO: 725645

Lat: 41.317N

Lon: 105.683W

Elev: 2215

StdP: 77.40

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24022

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-23.3	-19.6	-26.1	0.5	-22.1	-22.8	0.6	-17.8	17.8	1.0	15.8	0.2	2.6	250	0.779

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.8	29.6	12.8	28.3	12.5	27.0	12.2	15.2	23.8	14.5	23.1	13.8	22.5	6.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.7	12.0	16.7	11.8	11.3	16.4	10.8	10.6	15.9	50.0	23.6	47.6	22.9	45.5	22.2	19.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
15.0	12.9	11.7		-30.6	32.0	3.7	0.9	-33.3	32.7	-35.5	33.2	-37.6	33.7	-40.3	34.4
			DB	-30.5	16.9	3.3	0.8	-32.9	17.5	-34.8	17.9	-36.7	18.4	-39.1	19.0
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.5	-5.3	-4.2	0.1	3.4	8.5	14.6	18.5	17.2	12.6	5.7	-0.4	-5.6
	DBStd	9.84	6.25	6.30	5.67	4.86	4.41	3.59	2.26	2.32	4.00	5.44	6.41	6.63
	HDD10.0	2431	473	398	308	201	79	7	0	0	21	146	313	484
	HDD18.3	4739	732	632	566	447	303	118	25	47	173	391	563	743
	CDD10.0	784	0	0	1	3	34	146	263	224	99	14	1	0
	CDD18.3	51	0	0	0	0	0	6	30	13	2	0	0	0
	CDH23.3	1534	0	0	0	0	16	288	699	417	113	1	0	0
	CDH26.7	343	0	0	0	0	1	66	191	72	14	0	0	0
(14) Wind	WSAvg	5.4	5.9	6.1	6.0	6.0	5.5	5.2	4.4	4.4	4.7	5.4	5.6	5.8
(15) Precipitation	PrecAvg	274	10	10	18	26	37	36	36	29	26	20	15	10
	PrecMax	422	44	40	62	94	98	121	112	57	77	68	66	48
	PrecMin	137	1	0	0	5	0	2	4	10	1	0	0	1
	PrecStd	69	9	10	14	17	24	26	27	14	20	15	13	10
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.2	12.3	17.7	20.3	25.2	31.0	31.8	30.0	28.6	22.6	17.4	11.1
		MCWB	1.8	3.3	5.5	7.1	11.0	12.6	13.7	12.8	11.9	8.7	5.5	2.5
	2%	DB	7.5	9.1	14.8	18.1	22.8	28.6	30.2	28.5	26.2	20.3	14.3	7.6
		MCWB	1.0	1.7	4.3	6.1	9.8	11.7	13.3	12.6	11.1	7.9	4.4	1.2
	5%	DB	5.2	6.7	12.3	16.0	20.7	26.7	28.8	27.3	24.5	18.1	11.5	5.5
		MCWB	0.1	0.5	3.3	5.4	9.0	11.4	13.0	12.1	10.4	7.0	3.3	0.1
	10%	DB	3.2	4.5	9.5	13.3	18.3	24.7	27.4	26.0	22.6	15.6	8.8	3.5
		MCWB	-1.0	-0.3	2.2	4.6	8.2	11.1	12.8	12.1	9.9	6.1	2.1	-0.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.0	3.9	6.2	8.5	13.0	14.8	16.7	15.7	14.1	10.1	6.4	2.9
		MCDB	8.4	10.6	15.7	18.3	21.7	23.5	25.2	24.8	22.9	18.4	14.5	9.2
	2%	WB	1.6	2.2	4.8	6.9	11.0	13.8	15.6	14.7	12.8	8.9	4.9	1.6
		MCDB	6.3	8.0	13.7	15.4	19.0	23.1	24.2	23.3	21.5	17.9	13.3	7.0
	5%	WB	0.5	1.0	3.7	5.9	10.0	12.9	14.9	14.0	11.9	7.8	3.7	0.5
		MCDB	4.9	5.9	11.3	14.2	18.3	22.4	23.5	22.5	20.4	16.3	10.9	5.0
	10%	WB	-0.7	-0.1	2.5	5.0	9.0	12.0	14.2	13.4	11.0	6.7	2.4	-0.7
		MCDB	3.2	4.3	9.0	12.9	16.9	21.5	22.9	21.7	19.2	14.5	8.4	3.3
(35) Mean Daily Temperature Range	5% DB	MDBR	11.3	11.6	12.9	13.4	14.2	16.5	16.8	16.5	16.3	14.3	12.4	11.2
		MCDBR	12.4	13.4	16.7	18.4	18.4	20.0	19.3	18.9	19.6	18.2	15.5	12.3
		MCWBR	7.9	8.0	8.5	8.9	7.9	7.4	6.6	6.6	7.9	8.7	8.3	7.6
		MCDBR	11.7	12.1	15.5	16.6	16.4	17.3	16.0	15.6	16.1	16.3	14.7	11.7
	5% WB	MCWBR	7.8	7.6	8.2	8.4	7.6	7.3	6.1	6.0	6.8	8.0	8.1	7.5
(40) Clear-Sky Solar Irradiance	taub		0.220	0.230	0.255	0.277	0.297	0.301	0.329	0.326	0.285	0.254	0.234	0.225
	taud		2.541	2.527	2.518	2.501	2.514	2.516	2.471	2.467	2.581	2.644	2.596	2.527
	Ebn at Noon		977	1009	1009	998	978	970	938	934	961	969	951	940
	Edh at Noon		68	81	93	102	104	104	108	105	87	72	64	63
(44) All-Sky Solar Radiation	RadAvg		2.33	3.29	4.51	5.35	6.03	7.17	6.77	5.91	5.04	3.55	2.46	1.92
	RadStd		0.09	0.16	0.36	0.36	0.51	0.38	0.27	0.20	0.33	0.27	0.18	0.12

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (20 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page